Integration WS 3

Evaluate Q1-10

1. $\int \frac{x^2}{(x+1)(x+2)} dx$

2. $\int \frac{2x+3}{x^2+2x+5} dx$

3. $\int \frac{4}{x^2-2x-1} dx$

4. $\int \frac{4x-x^2}{(x+1)(x^2+4)} dx$

5. $\int \frac{10}{(x-1)(x^2+9)} dx$

6. $\int_0^2 \frac{x+1}{x^2+4} dx$

7. $\int_1^2 \frac{2x-3}{x^2-2x+2} dx$

8. $\int_0^3 \frac{x^2+4x+5}{(x+1)(x+3)} dx$

9. $\int_0^2 \frac{1+4x}{(4-x)(x^2+1)} dx$

10. $\int_0^{\sqrt{3}} \frac{8}{(x^2+1)(x^2+9)} dx$

11. a. Use the substitution $u = \frac{\pi}{4} - x$ to show that

$\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx = \int_0^{\frac{\pi}{4}} \ln\left[\frac{2}{1+\tan x}\right] dx$

b. Hence find the exact value of $\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx$

12. a. Show that $\int_{-a}^{a} f(x)dx = \int_0^a \{f(x) + f(-x)\}dx$

b. Hence deduce that $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x}{1+e^x} \sin^2 x\, dx = \frac{\pi}{4}$

13. a. Given that $I_{2n+1} = \int_0^1 x^{2n+1}e^{x^2}\, dx$, show that $I_{2n+1} = \frac{1}{2}e - nI_{2n-1}$

b. Hence or otherwise evaluate $\int_0^1 x^5 e^{x^2}\, dx$

14. a. If $I_n = \int x^3(\ln x)^n dx$, show that

$I_n = \frac{1}{4}[x^4(\ln x)^n - nI_{n-1}]$ for $n \geq 0$.

b. Find the exact value of $\int_1^2 x^3(\ln x)^2\, dx$.

15. Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x\, dx$ for $n = 0,1,2 \dots$

a. Show that $I_1 = \frac{1}{2} ln2$

b. Show that $I_n + I_{n-2} = \frac{1}{n-1}$ for $n = 2,3,4 \dots$

c. Explain why $I_n < I_{n-2}$, for $n \geq 2$, and deduce that $\frac{1}{2n+2} < I_n < \frac{1}{2n-2}$

d. By using the recurrence relation in park b, find I_5, and deduce that $\frac{2}{3} < \ln 2 < \frac{3}{4}$.

16. a. Show that $\int_0^{2a} f(x)dx = \int_0^a \{f(x) + f(2a-x)\}dx$.

b. Hence, or otherwise evaluate $\int_0^{\pi} \frac{xdx}{2+\sin x}$.

17. a. Evaluate $\int_0^{\frac{\pi}{4}} \sec\theta\, d\theta$

b. Hence, show that $\int_0^{\frac{\pi}{4}} \sec^3\theta\, d\theta = \frac{1}{2}[\sqrt{2} + \ln(\sqrt{2}+1)]$.

18. Given $I_n = \int_0^1 \frac{x^n}{x^2+1} dx$ for $n = 1,2,3, \dots$

a. Show that $I_n = \frac{1}{n-1} - I_{n-2}$ for $n \geq 2$.

b. Hence, evaluate $\int_0^1 \frac{x^5}{x^2+1} dx$.

19. a. Show that $\frac{1}{1+u^2} = 1 - u^2 + u^4 - u^6 + u^8 \dots$ and find range of u.

b. By integrating both sides of the above expression from 0 to x, show by suitable choice of x, that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots$

20. a. Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

b. Hence, or otherwise, calculate the value of $\int_0^{\pi} \frac{x\sin x}{2+\cos^2 x} dx$

21. If $I_n = \int x^n \sin x dx$, show that

a. $I_n + n(n-1)I_{n-2} = x^{n-1}(n\sin x - x\cos x)$

b. Hence, evaluate $\int_0^{\pi} x^2 \sin x\, dx$

22. Given that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$, show that $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x+\cos x} dx = \frac{\pi}{4}$

23. a. Show that $\sqrt{\frac{8-x}{x}} = \frac{4-x}{\sqrt{8x-x^2}} + \frac{4}{\sqrt{8x-x^2}}$

b. Hence evaluate $\int_0^2 \sqrt{\frac{8-x}{x}}\, dx$.

24. a. Given $f(x) = f(a-x)$ and using the substitution $u = a - x$, prove that $\int_0^a xf(x)dx = \frac{a}{2}\int_0^a f(x)dx$

b. Given that $f(x) = \frac{\sin x}{1+\cos^2 x}$, prove that $f(x) = f(\pi - x)$

c. Hence evaluate $\int_0^{\pi} \frac{x\sin x}{1+\cos^2 x} dx$.

25. Let $I_n = \int_0^1 x(1-x^5)^n dx$, where $n \geq 0$ is an integer

a. Show that $I_n = \frac{5n}{5n+2} I_{n-1}$ for $n \geq 1$.

b. Show that $I_n = \frac{5^n n!}{2\times7\times12\times\ldots\times(5n+2)}$

c. Hence evaluate I_4

26. a. Let $\mathrm{I}_n = \int_1^e (\ln x)^n\, dx$ and show that $\mathrm{I}_n = e - n\mathrm{I}_{n-1}$.

b. Hence evaluate I_3

27. Let $I_n = \int_0^1 \frac{x^{n-1}}{(x+1)^n} dx$, for $n = 1, 2, 3, \ldots$

a. Show that $\mathrm{I}_1 = \ln 2$

b. Use integration by parts to show that $\mathrm{I}_{n+1} = \mathrm{I}_n - \frac{1}{n2^n}$

c. The maximum value of $\frac{x}{x+1}$, for $0 \leq x \leq 1$, is $\frac{1}{2}$. Use this fact to show that $\mathrm{I}_{n+1} < \frac{1}{2}\mathrm{I}_n$.

d. Deduce that $\mathrm{I}_n < \frac{1}{n2^{n-1}}$.

e. Use the reduction formula in part (b) and the inequality in part (d) to show that $\frac{2}{3} < \ln 2 < \frac{17}{24}$

28. Given that $\int_0^{\pi} \frac{1}{5+3\cos x} dx = \frac{\pi}{4}$, show that $\int_0^{\pi} \frac{\cos x + 2\sin x}{5+3\cos x} dx = \frac{1}{12}(16\ln 2 - \pi)$

29. a. Use the substitution $u = t - t^{-1}$ to show that $\int \frac{1+t^2}{1+t^4} dt = \frac{1}{\sqrt{2}}\tan^{-1}\frac{\sqrt{2}(t^2-1)}{2t} + C$

b. Alternatively, use the result $(1+t^4) = (1+t^2)^2 - \left(\sqrt{2}t\right)^2$ and partial fractions to show that $\int \frac{1+t^2}{1+t^4} dt = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\sqrt{2}t}{1-t^2}\right) + C$

30. Consider the two new function $\cosh x = \frac{1}{2}(e^x + e^{-x})$ and $\sinh x = \frac{1}{2}(e^x - e^{-x})$.

Show that $\int_0^{\ln 2} \frac{1}{5\cosh x - 3\sinh x} dx = \frac{1}{2}\tan^{-1}\frac{1}{3}$

Answer

1. $x + \ln\left|\frac{x+1}{(x+2)^4}\right|$
2. $\ln(x^2+2x+5) + \frac{1}{2}\tan^{-1}\left(\frac{x+1}{2}\right)$
3. $\sqrt{2}\ln\left|\frac{x-1-\sqrt{2}}{x-1+\sqrt{2}}\right|$
4. $2\tan^{-1}\frac{x}{2} - \ln|x+1|$
5. $\ln\left|\frac{x-1}{\sqrt{x^2+9}}\right| - \frac{1}{3}\tan^{-1}\frac{x}{3}$
6. $\frac{1}{2}\ln 2 + \frac{\pi}{8}$
7. $\ln 2 - \frac{\pi}{4}$
8. $3 + \ln 2$
9. $\frac{1}{2}\ln 20$
10. $\frac{5\pi}{18}$
11. b. $\frac{\pi\ln 2}{8}$
13. b. $\frac{e}{2} - 1$
14. b. Let $I_2 = \int_1^2 x^3(\ln x)^2\, dx$, $I_2 = 4(\ln 2)^2 - 2ln2 + \frac{15}{32}$
15. c. $\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}$; d. $\frac{2}{3} < \ln 2 < \frac{3}{4}$
16. b. $\frac{\pi^2}{3\sqrt{3}}$
17. a. $\ln(\sqrt{2}+1)$
18. b. $\frac{1}{2}\ln 2 - \frac{1}{4}$
19. a. $-1 < u < 1$
20. b. $\frac{\pi}{\sqrt{2}} tan^{-1}\frac{1}{\sqrt{2}}$
21. b. $\pi^2 - 4$
23. b. $2\sqrt{3} + \frac{4\pi}{3}$
24. c. $\frac{\pi^2}{4}$
25. c. $I_4 = \frac{625}{2618}$
26. a. $6 - 2e$